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Review Article

Vestibulo-Cerebellar Modulation of Stress-Related Immune Responses: A Narrative Review

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Abstract

The vestibulo-cerebellum occupies a distinctive anatomical position, with connections to the hypothalamic-pituitary-adrenal (HPA) axis, the sympathetic-adrenomedullary (SAM) system, and the central noradrenergic system centred on the locus coeruleus. Because immune cells and lymphoid organs carry receptors for stress hormones and noradrenergic innervation, changes in these neuroendocrine pathways during stress can, in principle, alter immune function. A body of largely older, predominantly animal-based experimental work has examined whether vestibular and vestibulo-cerebellar stimulation can modulate stress-induced immune alterations, and this narrative review summarises that literature. We describe the physiology of the HPA and SAM stress axes and their known effects on immune cells; the neuroanatomical connections linking the vestibulo-cerebellum to the locus coeruleus and limbic-hypothalamic structures; proposed mechanisms of vestibular stimulation, including GABAergic modulation of anterior pituitary secretion and hippocampal involvement; and the preclinical and limited clinical evidence for vestibular neuro-immunomodulation. Vestibular stimulation is not a single, uniform intervention: rotational, galvanic, and caloric paradigms differ physiologically, and rotational stimulation in particular co-activates multiple sensory and autonomic pathways beyond the vestibulo-cerebellum itself, which complicates attribution of any observed immune effect to the vestibulo-cerebellum specifically. The available evidence is not sufficient to establish that vestibular stimulation reproducibly and specifically modulates immunity through the vestibulo-cerebellum in humans. We conclude that vestibular and vestibulo-cerebellar pathways may participate in the regulation of stress-related immune responses through interactions with neuroendocrine and autonomic systems, but that the precise neural circuitry, the relative contribution of individual pathways, and the translational relevance to humans remain incompletely defined, warranting well-designed experimental and clinical studies.

Key Words: HPA axis, Neuroimmunomodulation, Stress, Vestibular stimulation, Vestibulo-cerebellum

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I. Introduction: Stress Physiology and Immune Modulation

Since the time of Galen, it has been observed that mental state is linked to physical disease susceptibility[1]. A substantial literature now links psychosomatic and immune dysfunction to the body's physiological response to stress, mediated through homeostatic regulatory mechanisms whose chronic imbalance is reflected in somato-visceral dysfunction and disordered cognitive function[1].

Psychological stress is the neurological response to environmental demands that exceed an individual's coping capacity. The relationship between stress and immunity is more nuanced than a simple "stress suppresses immunity" framing suggests. A meta-analysis of three decades of inquiry found that acute stress can produce potentially adaptive changes in innate immunity — including lymphocytosis, mobilisation of cytotoxic lymphocyte subsets such as NK cells and CD8+ T cells, and increased NK-cell cytotoxicity — while chronic stress is more consistently associated with suppression of several immune measures[2]. More recent work similarly frames the neuroimmune response not as a unidirectional suppression pathway, but as an integrated response involving neural, endocrine, and immune mediators that can produce either activation/inflammation or suppression depending on the nature, duration, and context of the stressor[3].

Two neuroendocrine axes principally mediate this response. The hypothalamic-pituitary-adrenal (HPA) axis proceeds: hypothalamus → corticotropin-releasing hormone (CRH) → pituitary adrenocorticotropic hormone (ACTH) → adrenal glucocorticoids → immune cells. The sympathetic-adrenomedullary (SAM) system proceeds: sympathetic nervous system → noradrenaline/adrenaline → β-adrenergic receptors on immune cells. Table 1 summarises how the pattern of neuroendocrine activation differs by stress type and the immune consequences typically associated with each.

Stress type	**Typical neuroendocrine response**	**Potential immune consequence**
Acute stress	SNS/HPA activation	Redistribution/activation of selected immune cells
Repeated/short-term stress	Persistent neuroendocrine activation	Variable immune effects
Chronic stress	Prolonged HPA/SNS dysregulation	Immune suppression and/or chronic inflammation

Table 1: *Neuroendocrine and immune consequences by stress type (reproduced from reviewer feedback)*

Human evidence for stress-related immune modulation includes vaccine-response studies: medical students under examination stress showed reduced antibody and T-cell responses to hepatitis B vaccination, and caregivers under chronic stress produced lower IgG antibody titres to bacterial and meningococcal conjugate vaccines than controls, with social support identified as an ameliorating factor[3]. Chronic stress has also been associated with increased production of pro-inflammatory cytokines [3] which over prolonged inflammation poses a tissue-damage risk.

II. Neuroanatomy of the Vestibulo-Cerebellum and Its Connections

The vestibular nuclei send both excitatory and inhibitory projections to multiple brain regions; derangement of these circuits can alter sensorimotor behaviour. The vestibulo-cerebellum is anatomically positioned with links to the limbic-hypothalamic-striatal-diencephalic region and midbrain-reticular-vestibulomedullary structures, and in particular to the central noradrenergic system centred on the locus coeruleus[5,6]. Through these connections it is proposed to act as a neural switch mechanism integrating somato-visceral responses during stress, receiving signals conveyed by multiple neurotransmitters, peptides, and hormones[6].

Previous experimental studies have reported that lesions of the vestibulo-cerebellum produce measurable effects on immune function, including modulation of hypersensitivity-linked immune responses and humoral immunity, in animal models[7,8]. These findings should be read as animal, lesion-based experimental evidence rather than established human physiology; they motivate the mechanistic hypotheses discussed below without themselves demonstrating a clinically relevant effect in humans.

III. Mechanisms of Vestibular Stimulation

Vestibular stimulation is not a single, uniform intervention. Rotational stimulation, galvanic vestibular stimulation, caloric stimulation, and other vestibular paradigms differ in the receptors and pathways they engage and are understood to have distinct physiological effects; findings from one paradigm should not be generalised to the others without direct evidence.

Rotational stimulation at low frequency has been proposed as a non-invasive means of activating the vestibulo-cerebellum via the vestibular nuclei[6]. However, deterministic rotational motion — with its acceleration, deceleration, and abrupt-stoppage components — co-activates a wide range of additional receptors beyond the semicircular canals: the retina, ocular muscles, proprioceptors of the neck, face, body, and trunk, joint receptors, visceral interoceptors, and baroreceptors. This is an important conceptual limitation of the literature discussed here: **vestibular stimulation is not equivalent to selective vestibulo-cerebellar activation**, and rotational paradigms in particular cannot cleanly isolate a vestibulo-cerebellar-specific effect from the contribution of these co-activated sensory and autonomic pathways. Where the literature below reports an immune effect of "vestibular stimulation," this caveat should be assumed throughout. Specific stimulation parameters — frequency in Hz, angular velocity, acceleration/deceleration profile, and exposure duration — are not consistently reported in the sources reviewed here and are not reproduced in this review for that reason.

Two candidate mechanistic pathways have been proposed. First, vestibular stimulation has been reported to increase GABA release, which has separately been shown to modulate anterior pituitary hormone secretion [9]; whether this GABAergic pathway is the operative link between vestibular stimulation and HPA-axis inhibition specifically — as opposed to a parallel, unrelated observation — is not established by the cited evidence, and the underlying studies do not report whether the effect was blocked by GABA-receptor antagonism. Second, vestibular stimulation has been proposed to activate the hippocampal f tion, which in turn inhibits the HPA axis [10] Both pathways should be regarded as hypotheses requiring direct confirmation rather than established mechanisms.

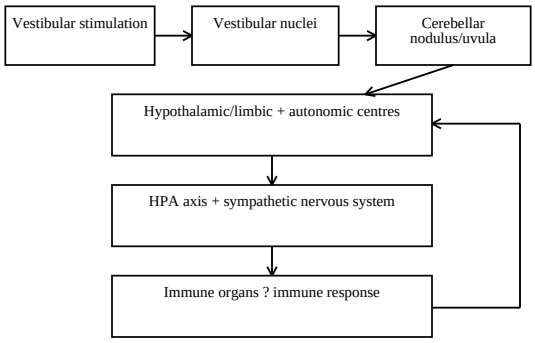
IV. Preclinical and Clinical Evidence of Vestibular Neuro-Immunomodulation

Neuroimmunomodulation is understood to involve close interaction between the endocrine, nervous, and immune systems. Several hormones and peptides — ACTH, growth hormone, endorphins, enkephalins, POMC-related

peptides, neuropeptide Y, and interleukin-1 — are synthesised by cells of both the neuroendocrine and immune systems, and extensive sympathetic innervation of lymphoid organs, with nerve endings and varicosities in close contact with immune cells, provides a plausible hard-wired route for neuronal regulation of immune function[11]. Broader signalling within this neuroendocrine-immune network has been described[12], including hor
pituitary cells in response to interleukin-1 13.

Directly relevant experimental work is limited in volume and age. An early rat study reported that restricted low-frequency vestibular stimulation corrected stress-induced alterations in immune status[14,15]. This finding is supportive of the review's central hypothesis but is a single, decades-old animal experiment; it does not by itself establish clinical efficacy or mechanism in humans, and should not be read as having done so.

A further source is cited for context on neu- odulation in brain disorders more broadly 16.



V. Methodological Note

This is a narrative, not a systematic, review. Sources were drawn from the author's prior publications on this topic and the general stress-neuroimmunology literature, rather than a systematic database search with defined inclusion criteria. Quantitative synthesis (meta-analysis) was not attempted, and the review does not claim comprehensive coverage of the field.

VI. Conclusion

Available experimental evidence suggests that vestibular and vestibulo-cerebellar pathways may participate in the regulation of stress-related immune responses through interactions with neuroendocrine and autonomic systems. However, the precise neural circuitry, the relative contribution of individual neuroendocrine pathways, and the translational relevance to humans remain incompletely defined. Well-designed experimental and clinical studies — particularly ones that isolate vestibulo-cerebellar activation from the other sensory and autonomic pathways co-activated by rotational stimulation — are required to determine whether vestibular stimulation has reproducible and clinically meaningful immunomodulatory effects.

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